SMART INDIA HACKATHON 2026

V.E.C.T.O.R

Velocity-Enhanced Clustering for Transactional Outlier Recognition

Problem Statement ID - SIH26146

Problem Statement Title - AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic

Theme - Blockchain & Cybersecurity

PS Category- Software

Team ID - 122958

Team Name - PROMETHEUSS

FROM TRANSACTION TRAFFIC TO INVESTIGATIVE LEADS

SMART INDIA HACKATHON 2025

INGEST → **PROFILE** → **CLUSTER** → **DETECT** → **PRIORITIZE**

Bitcoin transactions | Behavioral signatures | Related entities | Contextual anomalies | Explainable risk leads

VECTOR learns what "normal" looks like for each behavioral community - then surfaces deviations worth investigating.

THE CHALLENGE

Pseudonymous Wallets

Identity hidden behind addresses

Complex Fund Movement

Layering, rapid hopping, coordinated flows

Alert Overload

Hard to prioritize meaningful activity

Behavior-Aware

Detects deviations relative to an entity's behavioral community.

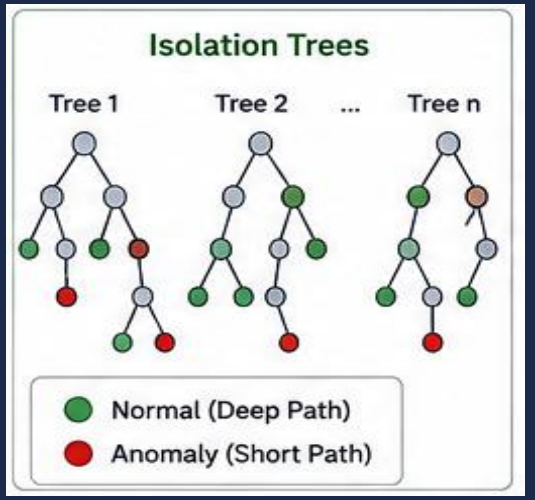
Adaptive

Continuous profiling and reclustering handles changing behavior.

Explainable

Every lead carries risk context, connected entities and evidence.

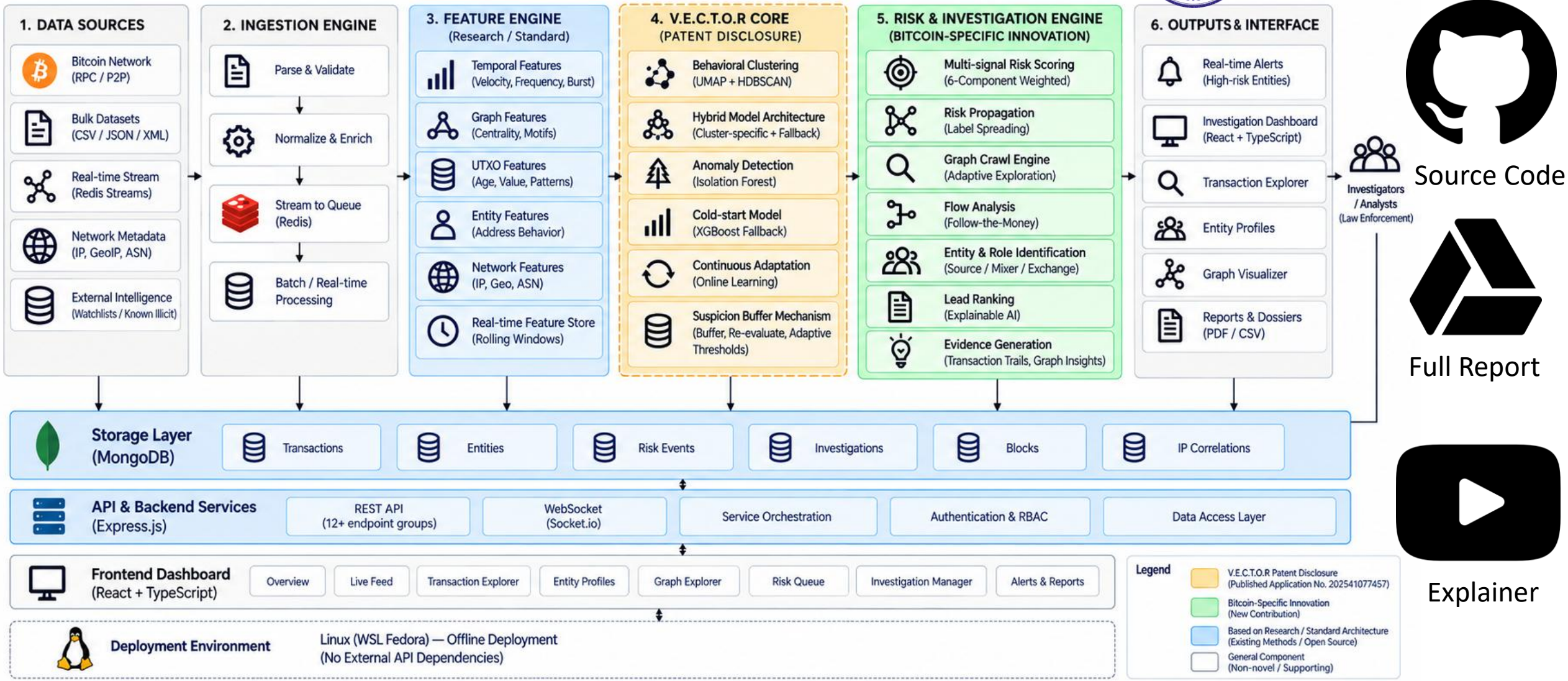
ANOMALY DETECTION USING ISOLATION FOREST



EXPLAINABLE INVESTIGATIVE LEAD

Risk Score	**HIGH (0.87)**
Connected Entities	**7**
Transaction Evidence	**23**
Investigation Priority	**P1**

TECHNICAL APPROACH



Source Code

Full Report

Explainer

FEASIBILITY AND VIABILITY

1 EXISTING VECTOR CORE

- **Real-time streaming**
  Redis Streams pipeline
- **ML scoring engine**
  behavioral feature scoring
- **Dynamic clustering**
  UMAP + HDBSCAN
- **Isolation Forest**
  cluster-specific detection
- **XGBoost fallback**
  cold-start / sparse entities
- **Live dashboard**
  real-time analytics & alerts

2 BITCOIN ADAPTATION

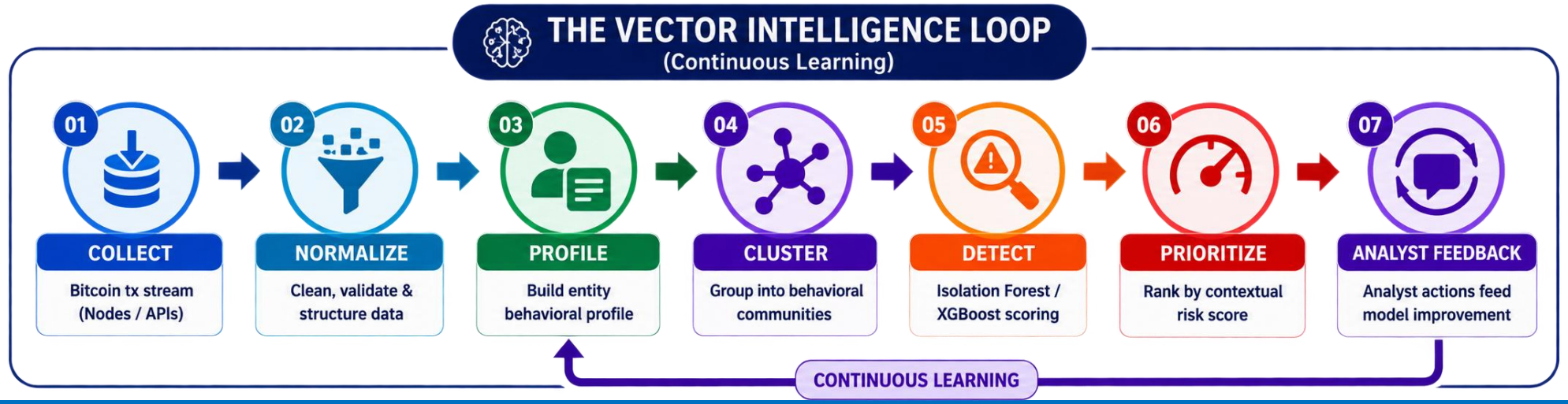
- **UTXO & transaction features**
  value, time, I/O, fees
- **Address / entity profiling**
  behavioral profile per entity
- **Behavioral communities**
  clusters addresses by behavior
- **Transaction-flow patterns**
  inflow/outflow, hops, chains
- **Temporal modeling**
  velocity, frequency, periodicity
- **Entity clustering & updating**
  reclusters as behavior evolves

3 NTRO DEPLOYMENT

- **Continuous monitoring**
  ingest, analyze & learn 24/7
- **Prioritized anomalies**
  contextual risk scoring
- **Evidence-backed leads**
  transactions, entities, patterns
- **Analyst investigation**
  deep dive with full context
- **Actionable intelligence**
  supports strategic decisions

CHALLENGES & MITIGATION

Challenge	Mitigation
Sparse / new entities	XGBoost cold-start model
Evolving behavior	Dynamic reclustering
False positives	Contextual cluster-specific detection
Analyst overload	Risk-based prioritization
Model opacity	Evidence-backed lead generation



IMPACT AND BENEFITS

MAJOR IMPACT

NATIONAL SECURITY

Earlier identification of suspicious crypto activity.

Continuous monitoring surfaces patterns that warrant deeper investigation.

LAW-ENFORCEMENT INTELLIGENCE

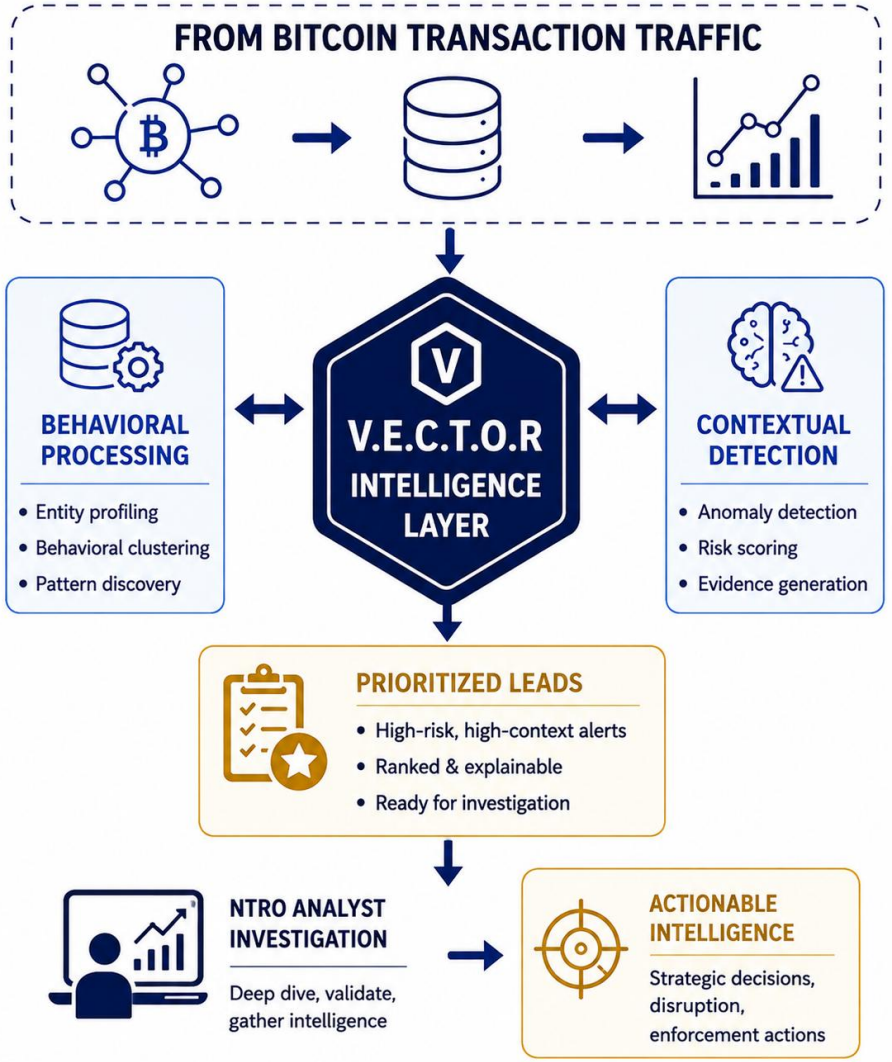
Evidence-backed investigative leads.

Connect transactions, entities and behavioral signals into a usable investigation context.

TECHNOLOGICAL IMPACT

Adaptive intelligence at scale.

Streaming architecture + behavioral clustering enables continuous analysis as transaction behavior evolves.



MAJOR BENEFITS

FASTER TRIAGE

Prioritize suspicious activity instead of reviewing every transaction.

Analysts spend time on what matters most.

HIGHER SIGNAL

Contextual anomaly detection reduces noise from isolated alerts.

More accurate leads with fewer false positives.

DEEPER VISIBILITY

Connect related addresses, entities and behavior.

Understand relationships, flows and communities behind transactions.

MID IMPACT

ECONOMIC IMPACT

Reduce investigative effort.

Move analysts from manual alert review to high-signal investigations, lowering time and resource cost.

RESEARCH AND REFERENCES

1. **Anomaly Detection in Bitcoin Network Using Unsupervised Learning Methods - Pham & Lee (2016)**
   Applied k-means, Mahalanobis distance and one-class SVM to Bitcoin transaction graphs, surfacing known theft/loss cases without labeled data - validates unsupervised clustering as a base for VECTOR.

2. **Anti-Money Laundering in Bitcoin: Graph Convolutional Networks - Weber et al. (2019)**
   Introduced the benchmark labeled Elliptic transaction graph and compared GCNs with random forests for illicit-transaction classification - informs VECTOR's feature design and evaluation.

3. **Real-Time Credit Card Fraud Detection using Streaming Analytics - Rajeshwari & Sathish Babu (2016)**
   Streaming analytics combined with Hidden Markov Models achieved 75–85% detection accuracy supports VECTOR's real-time behavioral scoring pipeline.

4. **Credit Card Fraud Detection using State-of-the-Art ML & DL - Alarfaj et al. (2022)**
   Found false positives a key limitation of existing models; gradient boosting emerged as the strongest ML approach - guided VECTOR's choice of XGBoost.

5. **Detecting Anomalies in Blockchain Transactions using ML Classifiers & Explainability - ScienceDirect (2024)**
   Ensemble classifiers combined with SHAP explainability improved both accuracy and analyst trust — directly supports VECTOR's explainable investigative leads.

6. **Clustering Algorithms for User & Entity Behavior Analytics — Artioli, Maci & Magrì (2024)**
   Benchmarked fifteen clustering algorithms for behavior analytics; HDBSCAN performed strongly on density-based behavioral grouping — validates VECTOR's clustering choice.

7. **Assessing Heuristic-Based Address Clustering for Bitcoin — Schnoering, Porthaux & Vazirgiannis (2024)**
   Evaluated heuristics for grouping Bitcoin addresses into real-world entities and introduced a clustering-ratio metric — informs VECTOR's entity-profiling step.

8. **Isolation Forest — Liu, Ting & Zhou (2008)**
   Introduced isolation-based anomaly detection, isolating anomalies through random partitioning instead of profiling normal points — the foundation of VECTOR's detection layer.

9. **XGBoost: A Scalable Tree Boosting System — Chen & Guestrin (2016)**
   Proposed a sparsity-aware, highly scalable gradient boosting framework — the basis for VECTOR's cold-start fallback model.

10. **Real-Time Fraud Detection in Digital Banking: Event Streaming Architecture — Mula (2025)**
    Combined Kafka, Redis and embedding-based anomaly detection for millisecond-level risk scoring — mirrors VECTOR's real-time streaming pipeline design.

Implementation Status — 10 October 2026

Current implementation and verification evidence

Repository scope

The checkout contains a React/TypeScript dashboard, Express REST and Socket.IO server, MongoDB persistence, and Python modules for Bitcoin transaction analysis. See README.md and System_Architecture.md for implemented boundaries.

Simulation data

Seeded Bitcoin records are synthetic fixtures marked SIMULATION. BITCOIN_SIMULATION_MODE=true reports simulation; no live Bitcoin Core node connection or cloud deployment is verified.

Verification completed

A fresh Python 3.12.14 environment installed requirements.txt and passed pip check. Backend/bitcoin tests passed: 60 tests and 7 subtests. One upstream XGBoost serialized-model compatibility warning remains; scikit-learn artifacts are pinned to 1.7.1. Frontend npm ci passed with zero reported vulnerabilities; five Node security tests passed. Lint has zero errors and three React Fast Refresh warnings. Production build passes; largest JS chunk is 304.36 kB (100.37 kB gzip) after route/tab splitting.

Local integration

The safe HTTP smoke passed six checks against a local app backed by temporary in-memory MongoDB 8.2.6. A guarded integration smoke passed one synthetic transaction through Express and the Python pipeline, retrieved it by txid from MongoDB, and observed its bitcoin:newTransaction Socket.IO event.

Still unverified

Durable or managed MongoDB recovery, Docker/Compose startup, cloud hosting, live Bitcoin Core connectivity, representative performance, and detection quality on an independent labeled set remain unverified. These checks do not establish production readiness.

Current readiness verdict: NO-GO

See AUDIT/GO_NO_GO.md and AUDIT/VALIDATION_RESULTS.md for the acceptance gate and remaining work. Earlier proposal mock values and historical audit counts are not current measured results.